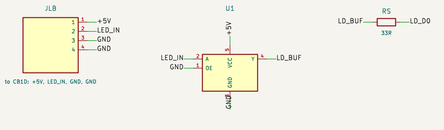
PF233-LB1: LB1 display: 66 x SK6812-MINI-E chain with 3V3 -> 5 V data buffer

How to read: parts sit in functional blocks (dashed frames); every pin ends in a short wire with its NET NAME: equal names are connected (also between blocks). A cross = intentionally unconnected pin.

Diode symbols point anode to cathode. Pin names are those of the library: items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

1 Data entry: 3V3 -> 5 V buffer

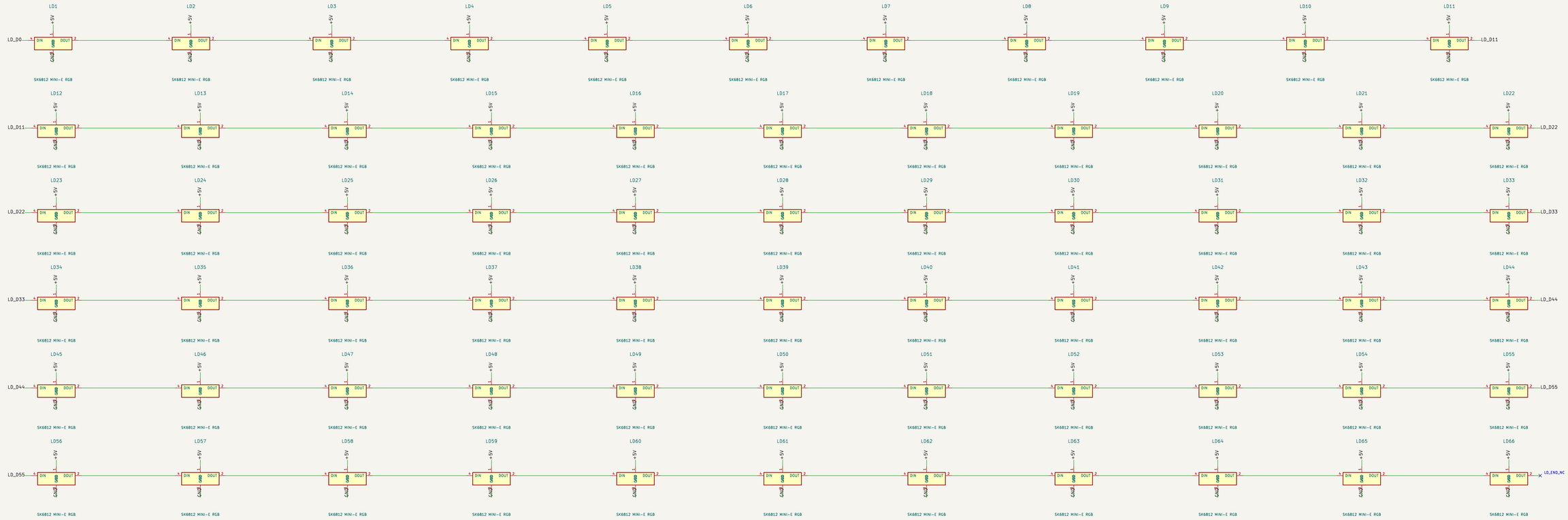
JLB: +5V, LED_IN, GND, GND from CB1D. U1 (74AHCT1G125) buffers LED_IN to 5 V logic; RS 33 R series damping -> LD_D0.



2 SK6812-MINI-E chain, 66 pixels

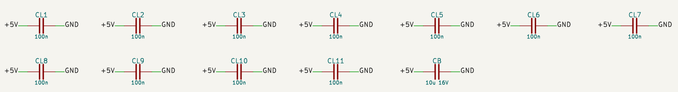
DOUT of each pixel goes straight to DIN of the next (wired); pins: 1 VDD (+5V), 2 DOUT, 3 GND, 4 DIN.

LD66 DOUT is left open (LD_END_NC).



3 Decoupling

CL1 to CL11: 100 nF across +5V / GND; CB 10 uF bulk.



Generated: src/pcb/sch_func.py: checked against the netlist.
MODELED design: not built, not bench-tested.
LB1 display: 66 x SK6812-MINI-E chain with 3V3 -> 5 V data buffer
ARX / PHI-FOLD 233

Sheet: /
File: PF233-LB1_functional.kicad_sch

Title: *PF233-LB1 functional schematic*

Size: User	Date: 2026-10-26	**Rev: r3**
KiCad E.D.A. 9.0.9		Id: 1/1